Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Azevedo II Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
30711 Pfitzer RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1977

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X069-X280-X013-XXXX X069-X280-X017-XXXX X069-X280-X031-XXXX

B. OPERATORS

Azevedo II, Frank

Operator name: Azevedo II, Frank

Telephone no.: (209) 854-6594 Landline (209) 495-1024 Cellular

Mailing Address Number and Street	City	State	Zip Code
30711 Pfitzer RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Azevedo II, Frank

Legal owner name: Azevedo II, Frank

Telephone no.: (209) 854-6594 Landline (209) 495-1024 Cellular

Mailing Address Number and Street	City	State	Zip Code
30711 Pfitzer RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	90	200	200	80	0
Number under roof	600	0	0	0	0	0
Maximum number	600	90	200	200	80	0
Average number	600	90	200	200	80	0
Avg live weight (lbs)	1,400	1,500	900	650		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,028.78 *tons per reporting period*

Total nitrogen from manure: 261,542.41 *lbs per reporting period*

After ammonia losses (30% loss applied): 183,079.69 *lbs per reporting period*

Total phosphorus from manure: 43,286.45 *lbs per reporting period*

Total potassium from manure: 115,693.94 *lbs per reporting period*

Total salt from manure: 303,205.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,448,500 *gallons*

Total nitrogen generated: 49,097.18 *lbs*

Total phosphorus generated: 2,937.00 *lbs*

Total potassium generated: 49,982.29 *lbs*

Total salt generated: 400,340.19 *lbs*

	8,448,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,448,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

Source Description

Source Description
Tile Drain

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
02/17/2023	Solid commercial fertilizer *8.7-0-23.2*	12.03 *ton*	Dry-weight	0.1	8.700000	0.000000	23.200000	0.000000
04/28/2023	Solid commercial fertilizer *6.3-25.2-25.2*	16.92 *ton*	Dry-weight	0.1	15.340000	61.360000	61.360000	0.000000
05/01/2023	Solid commercial fertilizer *4.7-10.3-4.2*	5.76 *ton*	Dry-weight	0.1	4.700000	10.300000	4.200000	0.000000
06/12/2023	Solid commercial fertilizer *UN32*	21.92 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/28/2023	Solid commercial fertilizer *27-0-0*	13.08 *ton*	Dry-weight	0.1	27.000000	0.000000	0.000000	0.000000
11/28/2023	Solid commercial fertilizer *4.6-0-27.7*	17.16 *ton*	Dry-weight	0.1	4.600000	0.000000	27.700000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	30,465.94	21,928.83	36,300.29	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	30,465.94	21,928.83	36,300.29	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	50	50	1	process wastewater	X069-X280-X018-XXXX X069-X280-X019-XXXX
Field 10	35	35	1	none	X069-X280-X033-XXXX
Field 11	54	54	1	none	X069-X280-X020-XXXX X069-X280-X023-XXXX X069-X280-X024-XXXX
Field 2	76	76	1	both	X069-X280-X032-XXXX
Field 3	36	36	2	process wastewater	X069-X280-X030-XXXX X069-X280-X031-XXXX
Field 4	44	44	2	both	X063-X240-X002-XXXX X069-X240-X024-XXXX
Field 5	31	31	2	both	X063-X240-X023-XXXX
Field 6A	67	67	1	process wastewater	X069-X280-X044-XXXX
Field 6B	51	51	1	none	X069-X280-X044-XXXX
Field 7-8	84	84	1	none	X063-X190-X041-XXXX X063-X190-X042-XXXX
Field 9	72	72	2	manure	X069-X280-X050-XXXX X069-X280-X051-XXXX X069-X280-X052-XXXX
Totals for areas that were used for application	376	376	11		
Totals for areas that were not used for application	224	224	4		
Land application area totals	600	600	15		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

Field 1

11/24/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 50 Plant date: 11/24/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	238.00 *ton*	As-is		10.6	27,800.00	2,230.00	17,100.00		9.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	4.76	264.66	21.23	162.79	781.30

Field 10

Field name: Field 10

11/28/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 35 Plant date: 11/28/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	181.00 *ton*	As-is		7.8	31,100.00	2,860.00	21,900.00		8.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	5.17	321.66	29.58	226.51	846.81

Field 11

Field name: Field 11

Field 11

12/14/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 54 Plant date: 12/14/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	286.00 *ton*	As-is		9.4	30,200.00	2,540.00	14,200.00		8.87

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	5.30	319.90	26.91	150.41	851.24

Field 2

Field name: Field 2

11/25/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 76 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	550.00 *ton*	As-is		6.8	27,300.00	2,670.00	19,900.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.24	395.13	38.64	288.03	1,375.93

Field 3

Field name: Field 3

Field 3

10/26/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	601.00 *ton*	As-is		70.8	7,300.00	1,260.00	12,500.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	132.00	20.40	99.60	0.00
Total actual harvest content	16.69	243.74	42.07	417.36	1,189.45

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/21/2023	1,126.00 *ton*	As-is		69.7	3,970.00	850.00	4,810.00		6.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	31.28	248.35	53.17	300.89	1,158.11

Field 4

Field name: Field 4

10/31/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 44 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	691.00 *ton*	As-is		72.2	5,850.00	980.00	8,260.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	132.00	20.40	99.60	0.00
Total actual harvest content	15.70	183.74	30.78	259.44	1,039.08

Field 4

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	1,434.00 *ton*	As-is		68.5	4,220.00	600.00	3,290.00		5.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	32.59	275.07	39.11	214.45	1,094.37

Field 5

Field name: Field 5

10/31/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 31 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	555.00 *ton*	As-is		54.0	7,160.00	1,480.00	10,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	17.90	256.37	52.99	383.13	1,745.92

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	839.00 *ton*	As-is		65.8	3,800.00	750.00	3,080.00		5.52

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.06	205.69	40.60	166.72	1,021.87

Field 6A

Field name: Field 6A

11/21/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 67 Plant date: 11/21/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	372.00 *ton*	As-is		9.6	25,700.00	2,100.00	17,800.00		8.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	5.55	285.39	23.32	197.66	896.43

Field 6B

Field name: Field 6B

11/25/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 51 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	470.00 *ton*	As-is		9.6	25,700.00	2,100.00	17,800.00		8.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	9.22	473.69	38.71	328.08	1,487.91

Field 7-8

Field name: Field 7-8

Field 7-8

11/20/2019: Almond, in shell

Crop: Almond, in shell Acres planted: 84 Plant date: 11/20/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	436.00 *ton*	As-is		4.2	18,800.00	1,650.00	24,800.00		19.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	5.19	195.16	17.13	257.45	1,929.32

Field 9

Field name: Field 9

10/30/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 72 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	390.00 *ton*	As-is		16.6	24,000.00	2,160.00	19,900.00		8.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	5.42	260.00	23.40	215.58	800.50

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,899.00 *ton*	As-is		72.2	2,900.00	430.00	3,330.00		4.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.38	152.98	22.68	175.66	626.17

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/24/2018: Alfalfa, hay

Field name: Field 1

Crop: Alfalfa, hay

Plant date: 11/24/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
6.3-25.2-25.2	Solid commercial fertilizer	15.34	61.36	61.36	0.00	
Application event totals		15.34	61.36	61.36	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	219.20	8,473,400.00 *gal*
Application event totals		0.00	0.00	0.00	219.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.33	1.47	49.75	237.93	360,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	225.10	8,701,530.00 *gal*
Application event totals		35.33	1.47	49.75	463.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	224.26	8,668,940.00 *gal*
Application event totals		0.00	0.00	0.00	224.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	219.20	8,473,400.00 *gal*
Application event totals		0.00	0.00	0.00	219.20	

Field 1 - 11/24/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	226.79	8,766,710.00 *gal*
Application event totals		0.00	0.00	0.00	226.79	

Field 10 - 11/28/2019: Alfalfa, hay

Field name: Field 10

Crop: Alfalfa, hay

Plant date: 11/28/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
6.3-25.2-25.2	Solid commercial fertilizer	15.34	61.36	61.36	0.00	
Application event totals		15.34	61.36	61.36	0.00	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	189.09	5,116,630.00 *gal*
Application event totals		0.00	0.00	0.00	189.09	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	297.49	8,049,730.00 *gal*
Application event totals		0.00	0.00	0.00	297.49	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	370.96	10,037,720.00 *gal*
Application event totals		0.00	0.00	0.00	370.96	

Field 10 - 11/28/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	199.93	5,409,940.00 *gal*
Application event totals		0.00	0.00	0.00	199.93	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	317.96	8,603,760.00 *gal*
Application event totals		0.00	0.00	0.00	317.96	

Field 11 - 12/14/2019: Alfalfa, hay

Field name: Field 11

Crop: Alfalfa, hay

Plant date: 12/14/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
6.3-25.2-25.2	Solid commercial fertilizer	15.34	61.36	61.36	0.00	
Application event totals		15.34	61.36	61.36	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	296.64	12,384,200.00 *gal*
Application event totals		0.00	0.00	0.00	296.64	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	302.11	12,612,330.00 *gal*
Application event totals		0.00	0.00	0.00	302.11	

Field 11 - 12/14/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	327.87	13,687,800.00 *gal*
Application event totals		0.00	0.00	0.00	327.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	249.02	10,396,210.00 *gal*
Application event totals		0.00	0.00	0.00	249.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	188.13	7,854,190.00 *gal*
Application event totals		0.00	0.00	0.00	188.13	

Field 2 - 11/25/2022: Alfalfa, hay

Field name: Field 2

Crop: Alfalfa, hay

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	409.09	168.16	651.06	6,214.62	750.00 *ton*
Application event totals		409.09	168.16	651.06	6,214.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.23	2.25	76.37	365.25	840,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	289.53	17,011,980.00 *gal*
Application event totals		54.23	2.25	76.37	654.78	

Field 2 - 11/25/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	329.47	19,358,460.00 *gal*
Application event totals		0.00	0.00	0.00	329.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	277.33	16,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	277.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	261.25	15,349,890.00 *gal*
Application event totals		0.00	0.00	0.00	261.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	242.94	14,274,420.00 *gal*
Application event totals		0.00	0.00	0.00	242.94	

Field 3 - 10/26/2022: Wheat, silage, soft dough

Field name: Field 3

Crop: Wheat, silage, soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.38	4.67	26.70	1,824.77	960,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	145.84	1,271,010.00 *gal*
Application event totals		110.38	4.67	26.70	1,970.61	

Field 3 - 10/26/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.38	4.67	26.70	1,824.77	960,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	329.08	2,867,920.00 *gal*
Application event totals		110.38	4.67	26.70	2,153.85	

Field 3 - 05/01/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4.7-10.3-4.2	Solid commercial fertilizer	2.96	6.48	2.64	0.00	
Application event totals		2.96	6.48	2.64	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.67	5.09	172.74	826.16	900,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	275.17	7,658,650.00 *gal*
Application event totals		122.67	5.09	172.74	1,101.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	101.87	2,835,330.00 *gal*
Application event totals		0.00	0.00	0.00	101.87	

Field 3 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.14	4.07	138.19	660.92	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	100.70	2,802,740.00 *gal*
Application event totals		98.14	4.07	138.19	761.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.29	5.36	137.19	477.33	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	137.00	3,813,030.00 *gal*
Application event totals		88.29	5.36	137.19	614.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.18	5,181,810.00 *gal*
Application event totals		0.00	0.00	0.00	186.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.43	1.79	45.73	159.11	240,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	138.17	3,845,620.00 *gal*
Application event totals		29.43	1.79	45.73	297.28	

Field 4 - 10/31/2022: Wheat, silage, soft dough

Field name: Field 4

Crop: Wheat, silage, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 10/31/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	544.61	5,801,020.00 *gal*
Application event totals		0.00	0.00	0.00	544.61	

12/07/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	177.33	15.73	141.33	1,167.97	703,000.00 *gal*
Application event totals		177.33	15.73	141.33	1,167.97	

Field 4 - 05/01/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	257.71	58.24	176.55	2,552.02	260.00 *ton*
Application event totals		257.71	58.24	176.55	2,552.02	

05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4.7-10.3-4.2	Solid commercial fertilizer	2.96	6.48	2.64	0.00	
Application event totals		2.96	6.48	2.64	0.00	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	629.44	21,411,630.00 *gal*
Application event totals		0.00	0.00	0.00	629.44	

Field 4 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	45.76	0.00	0.00	0.00	
Application event totals		45.76	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	332.44	11,308,730.00 *gal*
Application event totals		0.00	0.00	0.00	332.44	

06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
27-0-0	Solid commercial fertilizer	37.13	0.00	0.00	0.00	
Application event totals		37.13	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	240.47	8,180,090.00 *gal*
Application event totals		0.00	0.00	0.00	240.47	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	193.53	6,583,180.00 *gal*
Application event totals		0.00	0.00	0.00	193.53	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	114.97	3,910,800.00 *gal*
Application event totals		0.00	0.00	0.00	114.97	

Field 4 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	249.09	8,473,400.00 *gal*
Application event totals		0.00	0.00	0.00	249.09	

Field 5 - 10/31/2022: Wheat, silage, soft dough

Field name: Field 5

Crop: Wheat, silage, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	240.70	98.94	383.07	3,656.60	180.00 *ton*
Application event totals		240.70	98.94	383.07	3,656.60	

11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	564.54	4,236,700.00 *gal*
Application event totals		0.00	0.00	0.00	564.54	

12/07/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	81.63	7.24	65.06	537.65	228,000.00 *gal*
Application event totals		81.63	7.24	65.06	537.65	

Field 5 - 05/01/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/01/2023

Field 5 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	196.96	44.51	134.93	1,950.43	140.00 *ton*
Application event totals		196.96	44.51	134.93	1,950.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4.7-10.3-4.2	Solid commercial fertilizer	2.96	6.48	2.64	0.00	
Application event totals		2.96	6.48	2.64	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	354.91	8,505,990.00 *gal*
Application event totals		0.00	0.00	0.00	354.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	45.76	0.00	0.00	0.00	
Application event totals		45.76	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	323.64	7,756,420.00 *gal*
Application event totals		0.00	0.00	0.00	323.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
27-0-0	Solid commercial fertilizer	37.13	0.00	0.00	0.00	
Application event totals		37.13	0.00	0.00	0.00	

Field 5 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	254.29	6,094,330.00 *gal*
Application event totals		0.00	0.00	0.00	254.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	221.65	5,312,170.00 *gal*
Application event totals		0.00	0.00	0.00	221.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	271.96	6,518,000.00 *gal*
Application event totals		0.00	0.00	0.00	271.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	326.36	7,821,600.00 *gal*
Application event totals		0.00	0.00	0.00	326.36	

Field 6A - 11/21/2021: Alfalfa, hay

Field name: Field 6A

Crop: Alfalfa, hay

Plant date: 11/21/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	102.29	9.08	81.53	673.74	617,500.00 *gal*
Application event totals		102.29	9.08	81.53	673.74	

Field 6A - 11/21/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.71	10,396,210.00 *gal*
Application event totals		0.00	0.00	0.00	200.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.88	3.65	123.76	591.87	1,200,000.00 *gal*
Application event totals		87.88	3.65	123.76	591.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	201.33	10,428,800.00 *gal*
Application event totals		0.00	0.00	0.00	201.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	202.59	10,493,980.00 *gal*
Application event totals		0.00	0.00	0.00	202.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	183.09	9,483,690.00 *gal*
Application event totals		0.00	0.00	0.00	183.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	144.71	7,495,700.00 *gal*
Application event totals		0.00	0.00	0.00	144.71	

Field 6A - 11/21/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	211.40	10,950,240.00 *gal*
Application event totals		0.00	0.00	0.00	211.40	

Field 6B - 11/25/2022: Alfalfa, hay

Field name: Field 6B

Crop: Alfalfa, hay

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4.6-0-27.7	Solid commercial fertilizer	30.96	0.00	186.45	0.00	
Application event totals		30.96	0.00	186.45	0.00	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	267.80	10,559,160.00 *gal*
Application event totals		0.00	0.00	0.00	267.80	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	137.21	5,409,940.00 *gal*
Application event totals		0.00	0.00	0.00	137.21	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	249.62	9,842,180.00 *gal*
Application event totals		0.00	0.00	0.00	249.62	

Field 6B - 11/25/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	232.26	9,157,790.00 *gal*
Application event totals		0.00	0.00	0.00	232.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	184.32	7,267,570.00 *gal*
Application event totals		0.00	0.00	0.00	184.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	269.46	10,624,340.00 *gal*
Application event totals		0.00	0.00	0.00	269.46	

Field 7-8 - 11/20/2019: Almond, in shell

Field name: Field 7-8

Crop: Almond, in shell

Plant date: 11/20/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8.1-0-23.2	Solid commercial fertilizer	24.93	0.00	66.47	0.00	
Application event totals		24.93	0.00	66.47	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	165.61	10,754,700.00 *gal*
Application event totals		0.00	0.00	0.00	165.61	

Field 7-8 - 11/20/2019: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	240.88	15,643,200.00 *gal*
Application event totals		0.00	0.00	0.00	240.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	266.98	17,337,880.00 *gal*
Application event totals		0.00	0.00	0.00	266.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	158.58	10,298,440.00 *gal*
Application event totals		0.00	0.00	0.00	158.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	129.47	8,408,220.00 *gal*
Application event totals		0.00	0.00	0.00	129.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	110.40	7,169,800.00 *gal*
Application event totals		0.00	0.00	0.00	110.40	

Field 9 - 10/30/2022: Wheat Hay

Field name: Field 9

Crop: Wheat Hay

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 9 - 10/30/2022: Wheat Hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	287.88	118.34	458.15	4,373.25	500.00 *ton*
Application event totals		287.88	118.34	458.15	4,373.25	

Field 9 - 06/30/2023: Corn, silage

Field name: Field 9

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	308.55	17,174,930.00 *gal*
Application event totals		0.00	0.00	0.00	308.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4.7-10.3-4.2	Solid commercial fertilizer	2.96	6.48	2.64	0.00	
Application event totals		2.96	6.48	2.64	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	147.20	0.00	0.00	0.00	
Application event totals		147.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	496.48	27,636,320.00 *gal*
Application event totals		0.00	0.00	0.00	496.48	

Field 9 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
27-0-0	Solid commercial fertilizer	59.40	0.00	0.00	0.00	
Application event totals		59.40	0.00	0.00	0.00	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	227.17	12,644,920.00 *gal*
Application event totals		0.00	0.00	0.00	227.17	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	292.74	16,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	292.74	

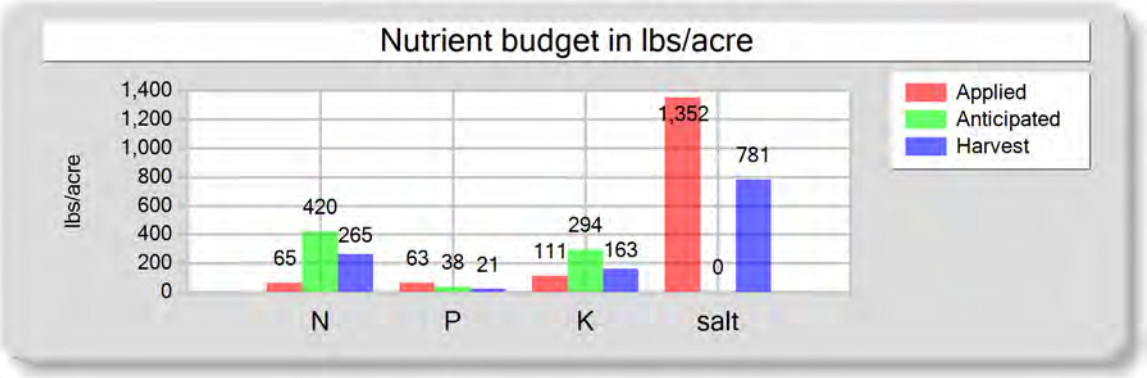
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	322.60	17,957,090.00 *gal*
Application event totals		0.00	0.00	0.00	322.60	

B. NUTRIENT BUDGET

Field 1 - 11/24/2018: Alfalfa, hay

Field name: Field 1 Crop: Alfalfa, hay Plant date: 11/24/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	15.34	61.36	61.36	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	35.33	1.47	49.75	237.93
Fresh water	0.00	0.00	0.00	1,114.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	64.67	62.83	111.11	1,352.49
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	264.66	21.23	162.79	781.30
Nutrient balance	-199.99	41.60	-51.68	571.19
Applied to removed ratio	0.24	2.96	0.68	1.73

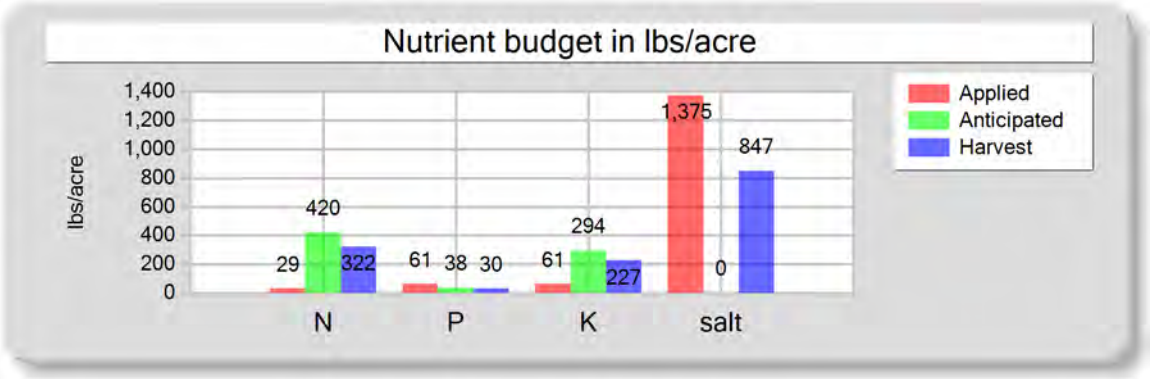
Fresh water applied	
43,083,980.00	*gallons*
1,586.64	*acre-inches*
31.73	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 10 - 11/28/2019: Alfalfa, hay

Field name: Field 10 Crop: Alfalfa, hay Plant date: 11/28/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	15.34	61.36	61.36	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,375.44
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	29.34	61.36	61.36	1,375.44
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	321.66	29.58	226.51	846.81
Nutrient balance	-292.32	31.78	-165.15	528.63
Applied to removed ratio	0.09	2.07	0.27	1.62

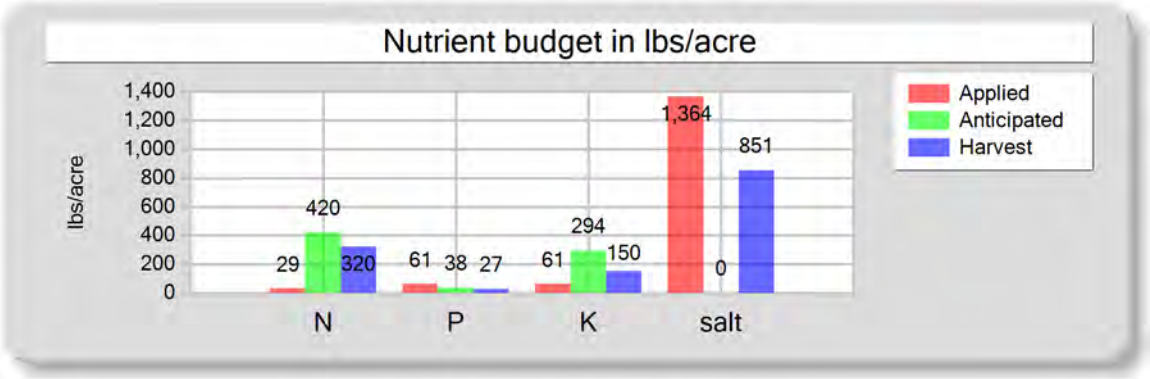
Fresh water applied	
37,217,780.00	*gallons*
1,370.60	*acre-inches*
39.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 12/14/2019: Alfalfa, hay

Field name: Field 11 Crop: Alfalfa, hay Plant date: 12/14/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	15.34	61.36	61.36	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,363.77
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	29.34	61.36	61.36	1,363.77
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	319.90	26.91	150.41	851.24
Nutrient balance	-290.56	34.45	-89.05	512.53
Applied to removed ratio	0.09	2.28	0.41	1.60

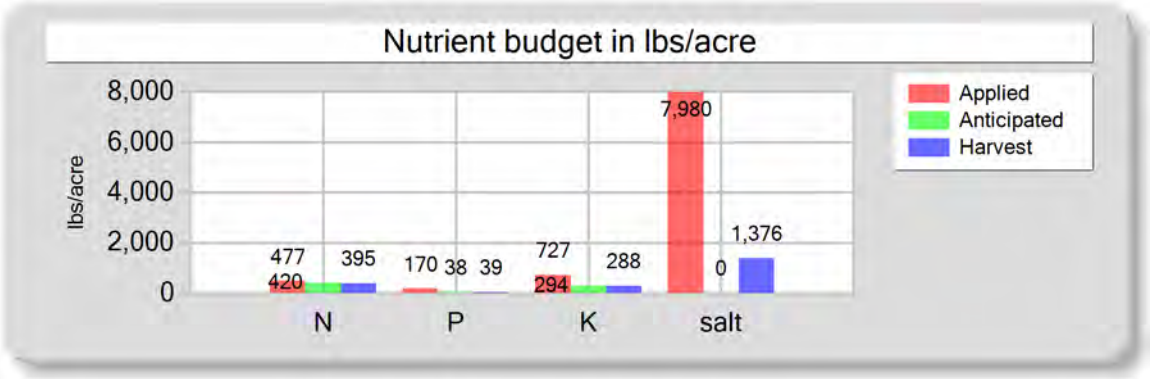
Fresh water applied	
56,934,730.00	*gallons*
2,096.71	*acre-inches*
38.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/25/2022: Alfalfa, hay

Field name: Field 2 Crop: Alfalfa, hay Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	409.09	168.16	651.06	6,214.62
Process wastewater	54.23	2.25	76.37	365.25
Fresh water	0.00	0.00	0.00	1,400.52
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	477.32	170.41	727.43	7,980.39
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	395.13	38.64	288.03	1,375.93
Nutrient balance	82.19	131.77	439.40	6,604.46
Applied to removed ratio	1.21	4.41	2.53	5.80

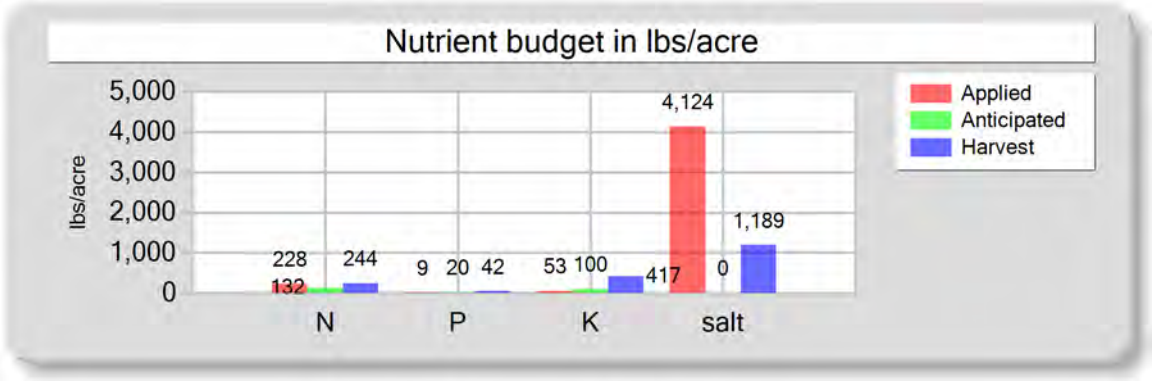
Fresh water applied	
82,289,750.00	*gallons*
3,030.45	*acre-inches*
39.87	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
0.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/26/2022: Wheat, silage, soft dough

Field name: Field 3 Crop: Wheat, silage, soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.75	9.35	53.41	3,649.55
Fresh water	0.00	0.00	0.00	474.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.75	9.35	53.41	4,124.46
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	243.74	42.07	417.36	1,189.45
Nutrient balance	-15.99	-32.72	-363.95	2,935.02
Applied to removed ratio	0.93	0.22	0.13	3.47

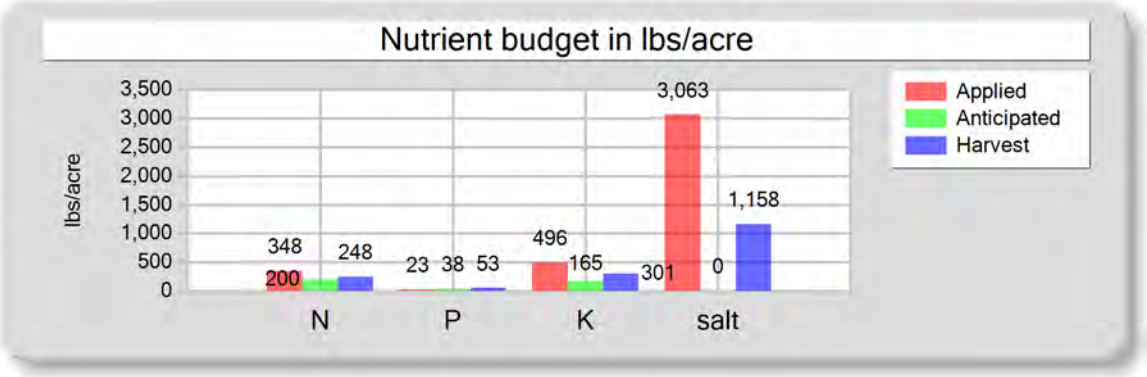
Fresh water applied	
4,138,930.00	*gallons*
152.42	*acre-inches*
4.23	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
1.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/01/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	2.96	6.48	2.64	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	338.53	16.31	493.86	2,123.52
Fresh water	0.00	0.00	0.00	939.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.49	22.79	496.50	3,062.63
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	248.35	53.17	300.89	1,158.11
Nutrient balance	100.14	-30.39	195.60	1,904.52
Applied to removed ratio	1.40	0.43	1.65	2.64

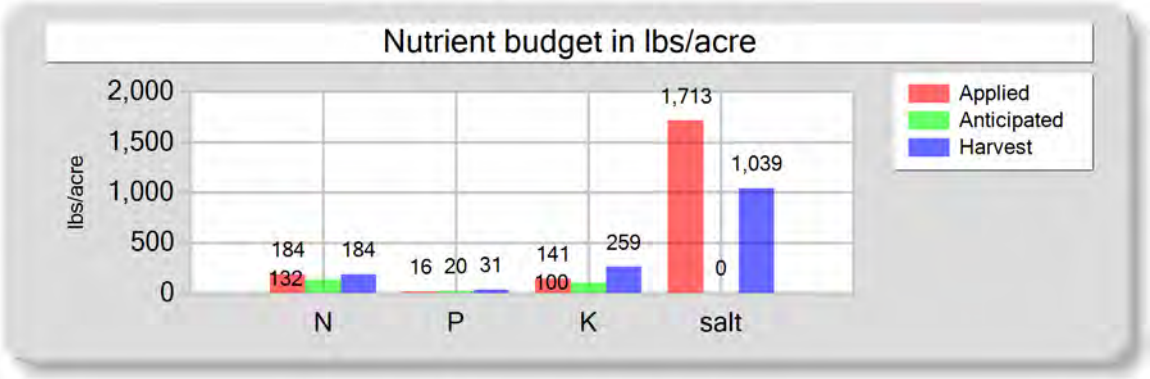
Fresh water applied	
26,137,180.00	*gallons*
962.54	*acre-inches*
26.74	*inches/acre*

Process wastewater applied	
2,580,000.00	*gallons*
95.01	*acre-inches*
2.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/31/2022: Wheat, silage, soft dough

Field name: Field 4 Crop: Wheat, silage, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	177.33	15.73	141.33	1,167.97
Fresh water	0.00	0.00	0.00	544.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.33	15.73	141.33	1,712.58
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	183.74	30.78	259.44	1,039.08
Nutrient balance	0.59	-15.05	-118.11	673.51
Applied to removed ratio	1.00	0.51	0.54	1.65

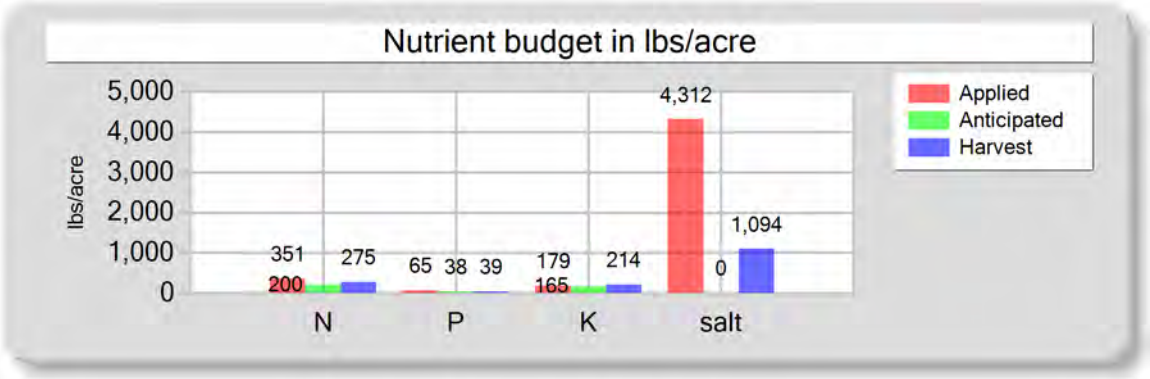
Fresh water applied	
5,801,020.00	*gallons*
213.63	*acre-inches*
4.86	*inches/acre*

Process wastewater applied	
703,000.00	*gallons*
25.89	*acre-inches*
0.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/01/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	85.85	6.48	2.64	0.00
Dry manure	257.71	58.24	176.55	2,552.02
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,759.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.56	64.72	179.19	4,311.96
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	275.07	39.11	214.45	1,094.37
Nutrient balance	75.49	25.61	-35.26	3,217.59
Applied to removed ratio	1.27	1.65	0.84	3.94

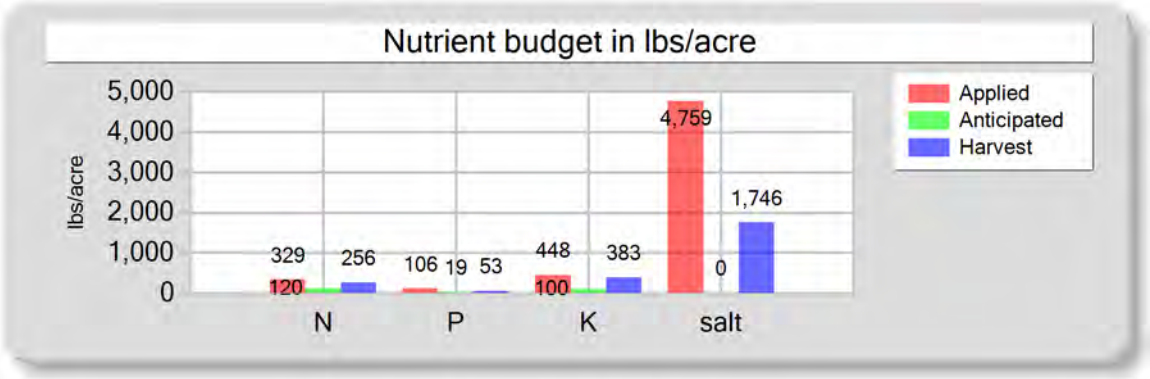
Fresh water applied	
59,867,830.00	*gallons*
2,204.73	*acre-inches*
50.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/31/2022: Wheat, silage, soft dough

Field name: Field 5 Crop: Wheat, silage, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	240.70	98.94	383.07	3,656.60
Process wastewater	81.63	7.24	65.06	537.65
Fresh water	0.00	0.00	0.00	564.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.33	106.19	448.13	4,758.80
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	256.37	52.99	383.13	1,745.92
Nutrient balance	72.96	53.19	65.00	3,012.88
Applied to removed ratio	1.28	2.00	1.17	2.73

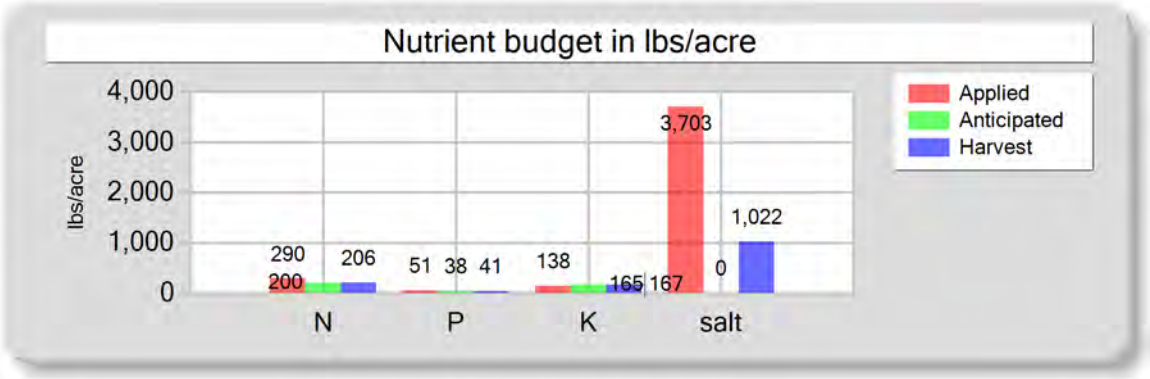
Fresh water applied	
4,236,700.00	*gallons*
156.02	*acre-inches*
5.03	*inches/acre*

Process wastewater applied	
228,000.00	*gallons*
8.40	*acre-inches*
0.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/01/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	85.85	6.48	2.64	0.00
Dry manure	196.96	44.51	134.93	1,950.43
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,752.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.81	50.99	137.57	3,703.23
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	205.69	40.60	166.72	1,021.87
Nutrient balance	84.12	10.39	-29.15	2,681.36
Applied to removed ratio	1.41	1.26	0.83	3.62

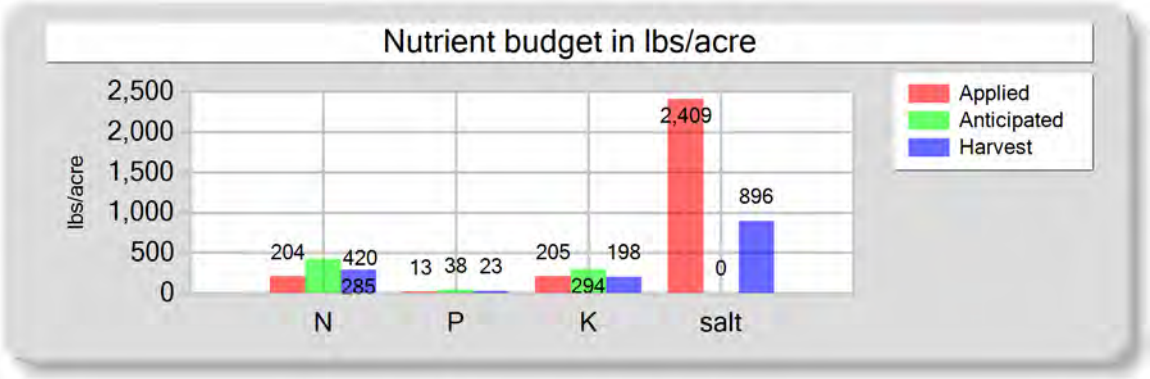
Fresh water applied	
42,008,510.00	*gallons*
1,547.03	*acre-inches*
49.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6A - 11/21/2021: Alfalfa, hay

Field name: Field 6A Crop: Alfalfa, hay Plant date: 11/21/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.18	12.72	205.28	1,265.61
Fresh water	0.00	0.00	0.00	1,143.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	204.18	12.72	205.28	2,409.44
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	285.39	23.32	197.66	896.43
Nutrient balance	-81.21	-10.60	7.62	1,513.01
Applied to removed ratio	0.72	0.55	1.04	2.69

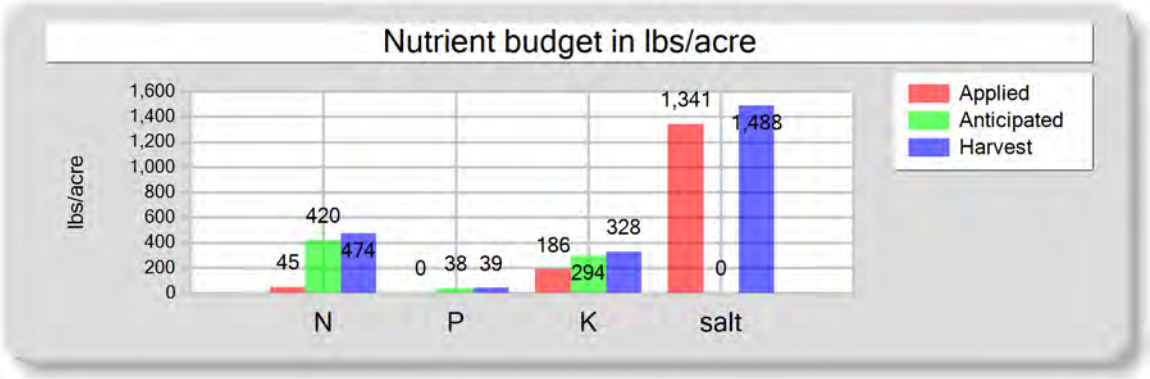
Fresh water applied	
59,248,620.00	*gallons*
2,181.93	*acre-inches*
32.57	*inches/acre*

Process wastewater applied	
1,817,500.00	*gallons*
66.93	*acre-inches*
1.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6B - 11/25/2022: Alfalfa, hay

Field name: Field 6B Crop: Alfalfa, hay Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.96	0.00	186.45	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,340.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	44.96	0.00	186.45	1,340.67
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	473.69	38.71	328.08	1,487.91
Nutrient balance	-428.73	-38.71	-141.63	-147.24
Applied to removed ratio	0.09	0.00	0.57	0.90

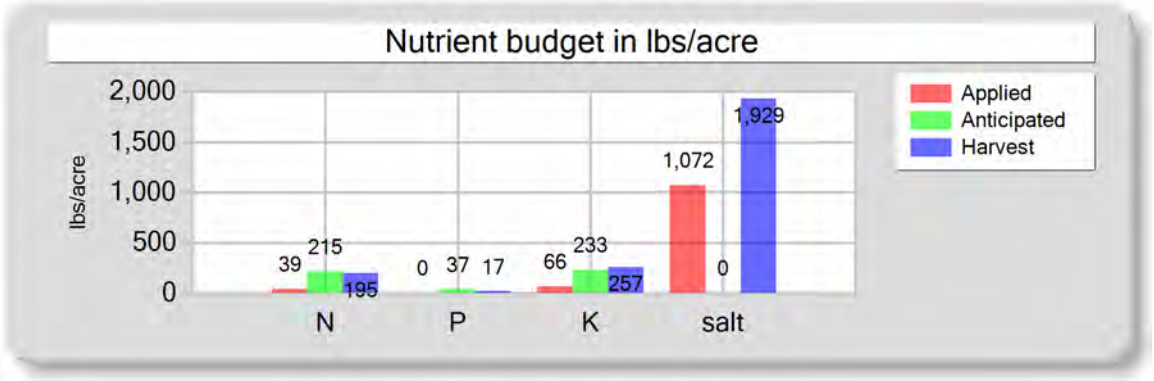
Fresh water applied	
52,860,980.00	*gallons*
1,946.69	*acre-inches*
38.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7-8 - 11/20/2019: Almond, in shell

Field name: Field 7-8 Crop: Almond, in shell Plant date: 11/20/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	24.93	0.00	66.47	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,071.92
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	38.93	0.00	66.47	1,071.92
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	195.16	17.13	257.45	1,929.32
Nutrient balance	-156.23	-17.13	-190.98	-857.40
Applied to removed ratio	0.20	0.00	0.26	0.56

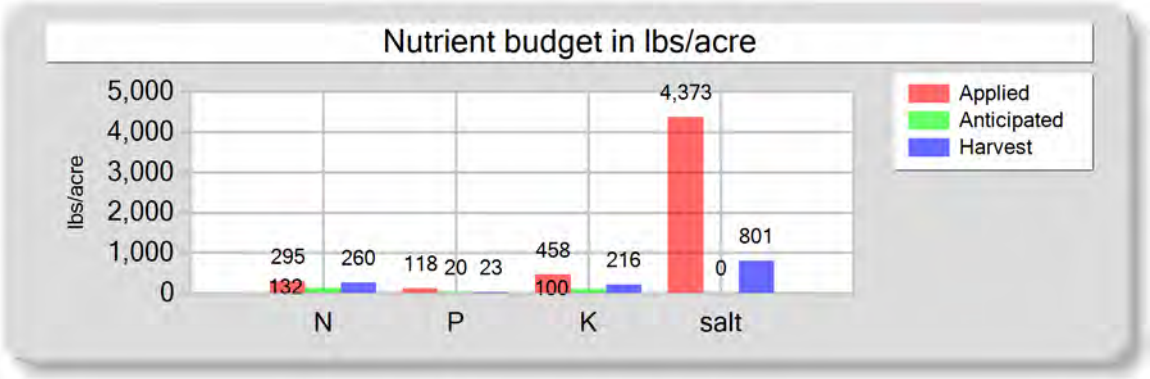
Fresh water applied	
69,612,240.00	*gallons*
2,563.58	*acre-inches*
30.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 10/30/2022: Wheat Hay

Field name: Field 9 Crop: Wheat Hay Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	287.88	118.34	458.15	4,373.25
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.88	118.34	458.15	4,373.25
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	260.00	23.40	215.58	800.50
Nutrient balance	34.88	94.94	242.57	3,572.75
Applied to removed ratio	1.13	5.06	2.13	5.46

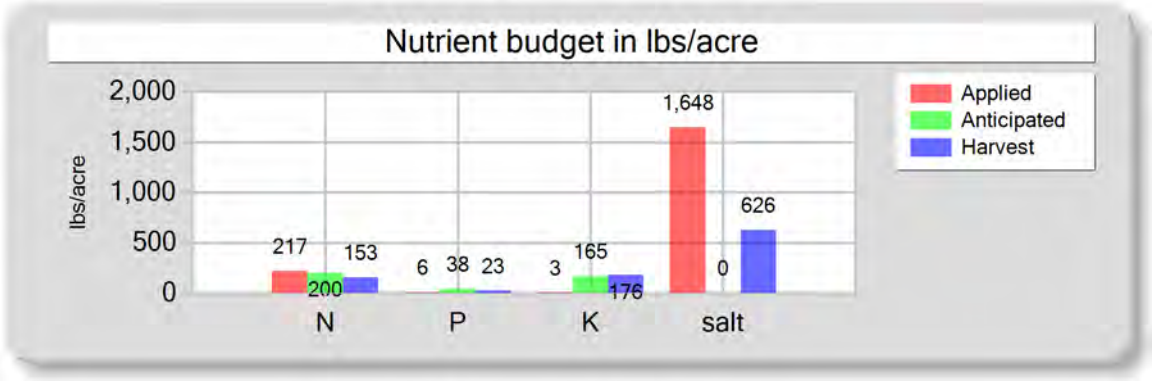
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 9 - 06/30/2023: Corn, silage

Field name: Field 9 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	209.56	6.48	2.64	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,647.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.56	6.48	2.64	1,647.53
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	152.98	22.68	175.66	626.17
Nutrient balance	63.59	-16.20	-173.02	1,021.36
Applied to removed ratio	1.42	0.29	0.02	2.63

Fresh water applied	
91,708,260.00	*gallons*
3,377.30	*acre-inches*
46.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure

Sample and source description: Manure

Sample date: 10/05/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	9,660.00	37,400.00	31,000.00	13,500.00	10,500.00	7,500.00	2,370.00		35.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Dry Manrue

Sample and source description: Dry Manrue

Sample date: 06/30/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 29.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,800.00	6,960.00	21,100.00							30.50
DL	100.00	100.00	100.00							0.01

Fall Manure

Sample and source description: Fall Manure

Sample date: 12/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	5,740.00	20,800.00							0.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

FalL Slurry

Sample and source description: FalL Slurry

Sample date: 12/07/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,330.00	350.00	0.00	0.00	118.00	1,060.00								8,400.00	8,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/07/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	496.00	21.00	0.00	0.00	21.00	120.00								8,330.00	8,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	577.00	375.00	0.00	0.00	89.40	829.00								7,830.00	6,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	434.00	0.00	0.00	24.40	828.00	123.00	152.00	498.00	3,980.00	0.00	235.00	96.10	8,070.00	3,960
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	529.00	384.00	0.00	0.00	32.10	822.00								7,370.00	2,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 12/27/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,200.00	630.00	0.00	0.00	116.00	735.00								7,380.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	652.00	454.00	0.00	0.00	53.40	878.00								8,200.00	8,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

CCID

CCID

CCID

Sample description: CCID

Sample date: 08/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,940.00	495
DL	0.50	0.50	0.50								1.00	10

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			250.00				
DL			4.00				

Field 10

Field 10

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			247.00				
DL			5.00				

Field 11

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			249.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			242.00				
DL			4.00				

Field 2

Field 2

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			278.00				
DL			4.00				

Field 3

Soil

Sample and source description: Soil

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			287.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			248.00				
DL			4.00				

Field 5

Field 5

Soil

Sample and source description: Soil

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			248.00				
DL			4.00				

Field 6A

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			250.00				
DL			4.00				

Field 6B

Soil

Sample and source description: Soil

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			251.00				
DL			4.00				

Field 7-8

Field 7-8

Soil

Sample and source description: Soil

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			256.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			264.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			251.00				
DL			4.00				

Field 9

Field 9

Soil

Sample and source description: Soil

Sample date: 06/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			245.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/24/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	2,230.00	17,100.00		9.18
DL	100.00	100.00	100.00		0.01

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,700.00	820.00	6,350.00		9.20
DL	100.00	100.00	100.00		0.01

Field 10 - 11/28/2019: Alfalfa, hay

Field 10 - 11/28/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/17/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,100.00	2,860.00	21,900.00		8.88
DL	100.00	100.00	100.00		0.01

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,820.00	920.00	5,680.00		9.19
DL	100.00	100.00	100.00		0.01

Field 11 - 12/14/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,200.00	2,540.00	14,200.00		8.87
DL	100.00	100.00	100.00		0.01

Field 11 - 12/14/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	1,180.00	7,870.00		10.00
DL	100.00	100.00	100.00		0.01

Field 2 - 11/25/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/17/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	2,670.00	19,900.00		10.20
DL	100.00	100.00	100.00		0.01

Field 3 - 10/26/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	1,260.00	12,500.00		12.20
DL	100.00	100.00	100.00		0.01

Field 3 - 05/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	850.00	4,810.00		6.11
DL	100.00	100.00	100.00		0.01

Field 4 - 10/31/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,850.00	980.00	8,260.00		11.90
DL	100.00	100.00	100.00		0.01

Field 4 - 05/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,220.00	600.00	3,290.00		5.33
DL	100.00	100.00	100.00		0.01

Field 5 - 10/31/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,160.00	1,480.00	10,700.00		10.60
DL	100.00	100.00	100.00		0.01

Field 5 - 05/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	750.00	3,080.00		5.52
DL	100.00	100.00	100.00		0.01

Field 6A - 11/21/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,700.00	2,100.00	17,800.00		8.93
DL	100.00	100.00	100.00		0.01

Field 6A - 11/21/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,610.00	970.00	8,600.00		11.70
DL	100.00	100.00	100.00		0.01

Field 6B - 11/25/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,700.00	2,100.00	17,800.00		8.93
DL	100.00	100.00	100.00		0.01

Field 7-8 - 11/20/2019: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 4.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	1,650.00	24,800.00		19.40
DL	100.00	100.00	100.00		0.01

Field 9 - 10/30/2022: Wheat Hay

Wheat Hay

Sample and source description: Wheat Hay

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	2,160.00	19,900.00		8.86
DL	100.00	100.00	100.00		0.01

Field 9 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 12/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,900.00	430.00	3,330.00		4.27
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

Tile Drain

Tile Drain

Sample description: Tile Drain

Sample date: 09/25/2023 Source of analysis: Lab analysis

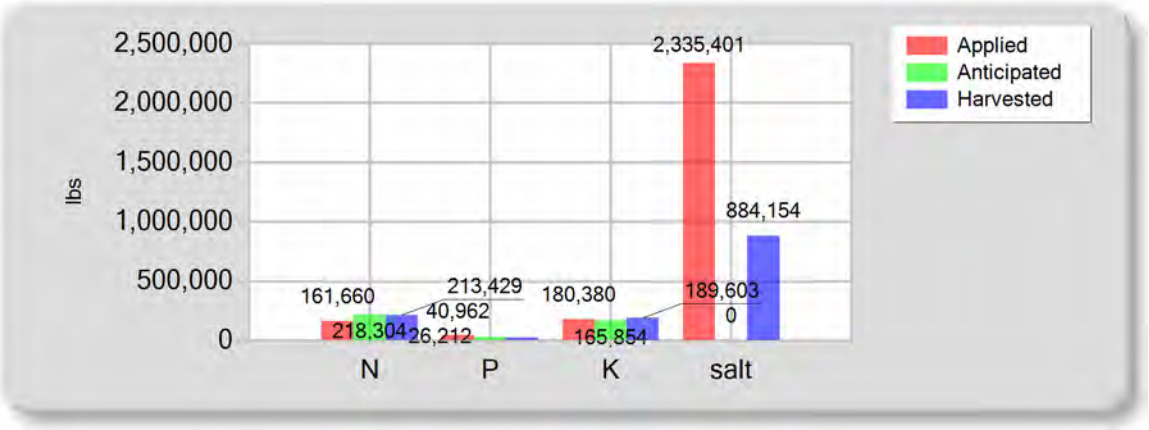
	NH4-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	6.21	0.00	1,790.00	940
DL	0.50	0.50	0.50	1.00	10

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

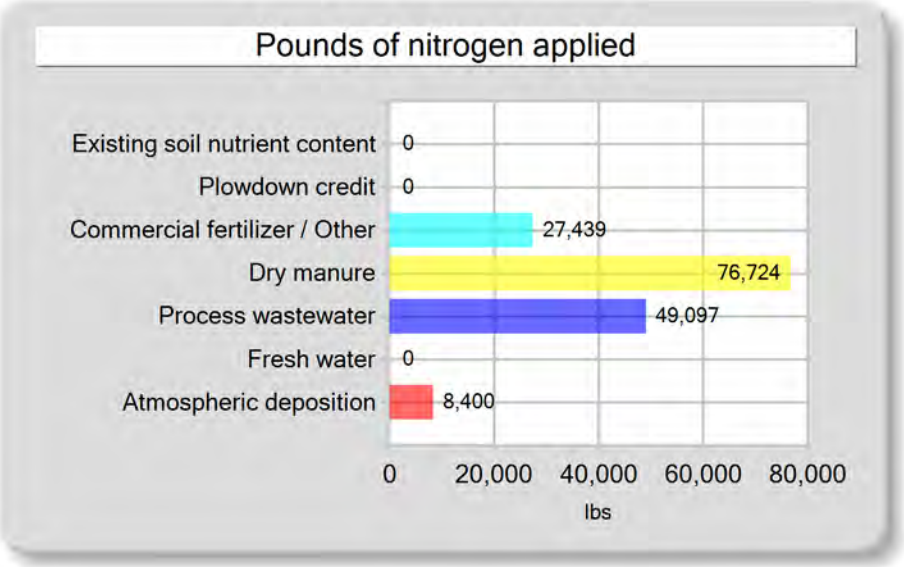
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

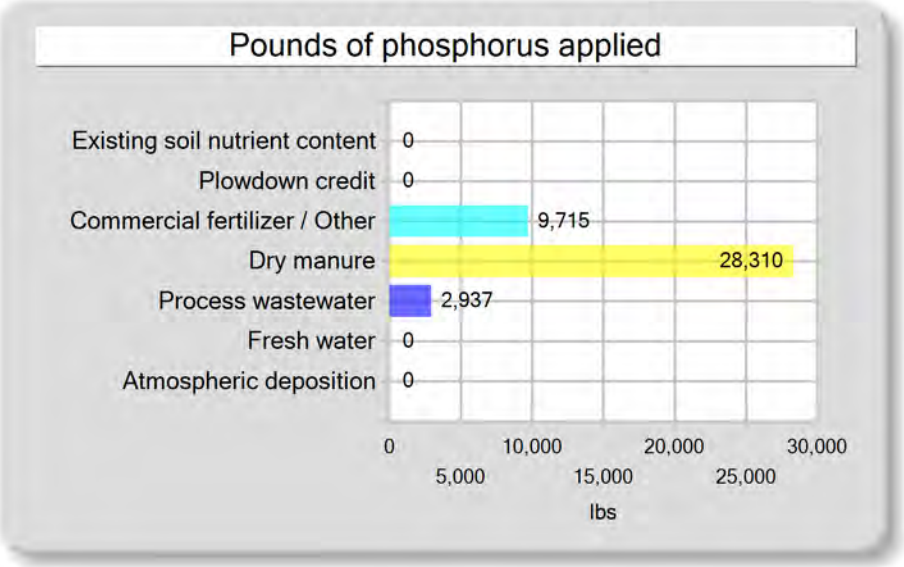
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	27,438.97	9,714.88	24,104.59	0.00
Dry manure	76,724.34	28,309.69	106,293.29	1,073,291.64
Process wastewater	49,097.18	2,937.00	49,982.29	400,340.19
Fresh water	0.00	0.00	0.00	861,769.54
Atmospheric deposition	8,400.00	0.00	0.00	0.00
Total nutrients applied	161,660.49	40,961.56	180,380.17	2,335,401.37
Anticipated crop nutrient removal	218,304.00	26,211.90	165,853.80	0.00
Actual crop nutrient removal	213,428.70	24,185.00	189,603.34	884,154.09
Nutrient balance	-51,768.21	16,776.56	-9,223.17	1,451,247.29
Applied to removed ratio	0.76	1.69	0.95	2.64

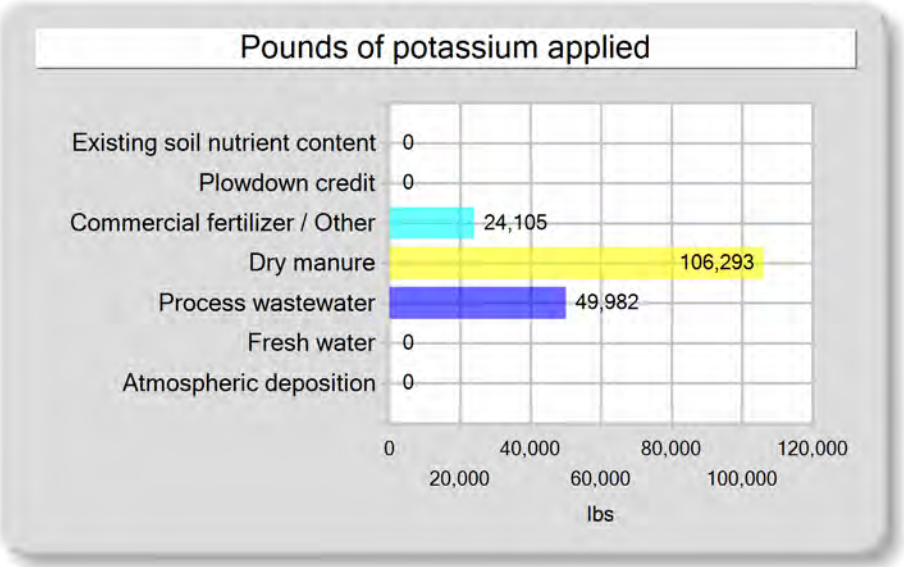
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

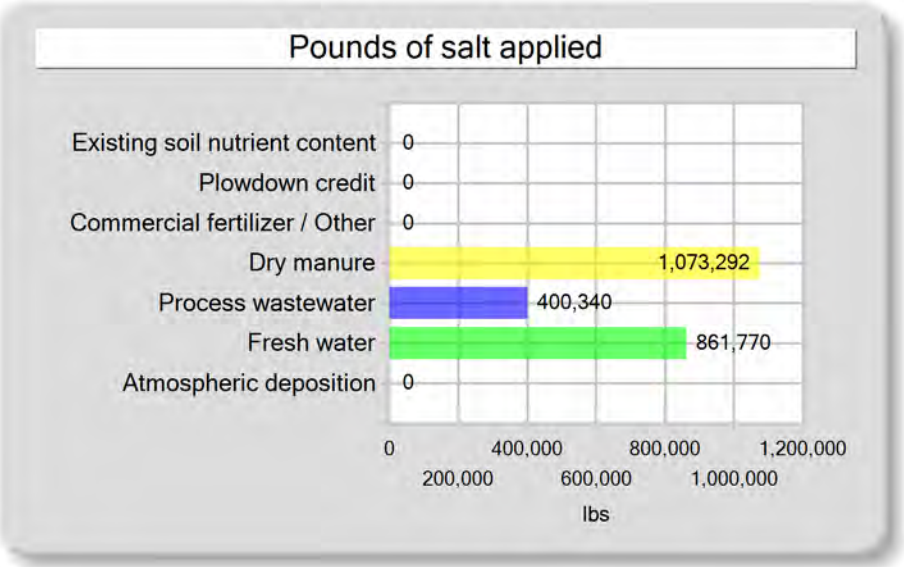


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Frank Azevedo II
Frank Azevedo II (Jun 11, 2024 09:00 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Azevedo II	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 11, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
FRANK AZEVEDO II
30711 Pfitzer Rd.
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/26/23 9:17 AM
SUBMITTED BY: Joe
ANALYZED DATE: 9/26/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
House Well	T3269010A	Electrical Conductivity (EC)	EPA 120.1 and	1.68	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1070	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.035
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
FRANK AZEVEDO II
30711 Pfitzer Rd.
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/26/23 9:17 AM
SUBMITTED BY: Joe
ANALYZED DATE: 9/26/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Pfitzer Well	T3269010B	Electrical Conductivity (EC)	EPA 120.1 and	1.29	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	824	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Electrical Conductivity (EC)	97.9			0.035
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42

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www.denelelabs.com

Compliance Analysis Report

Customer
FRANK AZEVEDO II
30711 Pfitzer Rd.
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/26/23 9:17 AM
SUBMITTED BY: Joe
ANALYZED DATE: 9/26/23

Tile Drain (H2)
SOURCE: Tile Drain

NOTES: Temperature was above qualifying at 9.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Tile Drain	T3269010C	Electrical Conductivity (EC)	EPA 120.1 and	1.79	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	940	mg/L
		Total Phosphorus (TP)	EPA 200.7	< 0.200	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.21	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1150	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	93.4 (T3269010A)	95.8 (T3269010A)	2.53
Nitrate Nitrogen (NO3-N)	98.2	98.9 (T3269039A)	91.9 (T3269039A)	7.42
Total Dissolved Solids (TDS)	96.0			5.20
Total Phosphorus (TP)	99.1	102 (W3284011A)	105 (W3284011A)	2.53

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Frank Azevedo !!
Mailing address: 30711 Pfitzer RD
(not facility address) Gustine CA 95322
Phone#:

Authorized Copy Released To: F&R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	9/25		House Well		10.0°		✓				
2	9/25		Pfitzer Well		10.4°		✓				
3	9/25		Tile Drain		9.8°						H2
4	9/25		WW Q3		11.6°			✓			
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged In By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	9-26-23 9:17			T3269010

2023 Frank Azevedo Dairy Annual Report

Final Audit Report 2024-06-11

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